TOPIC 1 Linear Equations and Inequalities

Name _____ Date _____

Reminders

Words: at least ≥ · at most ≤ · more than > · fewer than <.

Solve like an equation: undo + and −, then × and ÷.

Flip the sign when you multiply or divide both sides by a negative number.

Graph: ○ for < and >, ● for ≤ and ≥. Shade right for > and ≥, left for < and ≤.

1. What is the solution set for $3(4 - x) > -2x + 10$?

Tip: Distribute, move the x-terms, and flip if you divide by a negative.

A. $x < 2$

B. $x > 2$

C. $x \leq -2$

$12 - 3x > -2x + 10 \rightarrow -x > -2 \rightarrow x < 2$

2. Solve the inequality, and graph the solution on the number line.

$1.5m + 4 \geq -2(m - 9)$

> **Tip:** Distribute the −2 to both terms.

Distribute: 1.5m + 4 ≥ _____

Add 2m, subtract 4: _____ ≥ _____

m _____ _____

***m* ≥ 4, closed circle at 4, shaded right**

$1.5m + 4 \geq -2m + 18 \rightarrow 3.5m \geq 14 \rightarrow m \geq 4$

3. Solve the inequality.

$-2(x + 1) \geq 8$

> **Tip:** Divide by −2 at the end: flip.

Distribute: _____ ≥ 8

Add 2: −2x ≥ _____

Divide by −2 and flip: x _____ _____

***x* ≤ −5**

$-2x - 2 \geq 8 \rightarrow -2x \geq 10 \rightarrow x \leq -5$